

Chess Opening Trainer — Overview

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1 Chess Opening Trainer

1.1 What is it?

The Chess Opening Trainer (COT), is a cognitive assistant with one of its main purposes being to be able to play against a medium level human player effectively during the first stages of the game (the "Opening game", as it is usually referred to¹), thus the characterisation as a "trainer". Even if nowadays the chess opening theory has been largely, if not entirely, systematized and organized in large databases (for instance, *ECO — Encyclopedia of Chess Openings*, see [1]), implying that such an application could be based solely on a database and effective search therein, it is evident that this is not the way in which a human player plays during the opening of a game².

Instead of learning by heart dozens of different variations of openings, a medium-level human player would probably learn the first starting moves and try to get the basic ideas behind these moves, so as to be able to make a plan for their middle game, based on the characteristics of their position as they have been shaped by the opening moves. So, what the dominant element of the way a casual human player plays during opening is that they are using some *strategic* and *general purpose* rules, rather than simply searching throughout an opening database. This behaviour is the one that COT is intended to imitate, in the sense that it will be trained to be able to assess and use rules in order to evaluate its future moves and not merely search space search — more about its design will be discussed in the next sections.

¹Usually, the first 10-15 moves, before one enters the "Middle game". The last phase of the game is usually referred to as "Finale".

²With the modest claim that, at any time during the first 10 moves, each player has 4 meaningful possible moves to consider, this gives us a total of 4^{20} possible states to search throughout, which is roughly equal to:

$$4^{20} = 10^{20 \log_{10} 4} > 10^{10},$$

being way more than a casual human player would be capable of processing in a 10 moves analysis during the opening.

1.2 Motivation behind COT

As it has been made clear up to now, COT is intended to be used by mainly casual chess players so as to train themselves on chess opening theory, without having the sense that they are playing against a machine; that is, encountering too often the same positions on the board just because their opponent plays "by the book".

However, there is also another, more theoretic motivation behind the design of such a cognitive agent, which is to explore new ways one could study chess opening theory and cognitive knowledge acquisition regarding it. Up to now, the usual and expected way to deal with chess was, roughly, to consider the various positions occurring during a game as points in a search space and conduct efficient search in this space combining both state-of-the-art algorithms as well as immense databases so as to tackle the time complexity issues that occur. No matter how effective this procedure has been — and it has been absolutely effective, following Kasparov's 1997 defeat against Deep Blue — it still remains dubious that a human player plays so inhumanly, lacking any sense of intuition and being based solely on brute-force techniques.

On the contrary, in the approach adopted in COT, chess, and especially opening theory, is viewed under the scope of acting according to some specific set of rules that are commonly accepted. It is expected that this reliance on empirical rules will, up to some point, outweigh the reliance on depth analysis and search in the search space of the following positions. The aforementioned view also reveals the way empirical rules are viewed under the COT framework; as a collection of human experience on the opening theory.

According to this view, one may consider that another motive for implementing an approach like COT in chess is that of exploring how one verifies already existing knowledge and believes (the rules), instead of exploring new possibilities (the procedure of searching). It is thought that such an approach may not be as efficient as an exhaustive search, it is expected, however, to produce adequate results with minimal computational cost, leaving, thus, a lot of space for improvement.

1.3 Target groups

COT is evidently designed for intermediate level players who would like to practice their opening techniques. However, apart from them, COT is also designed for whoever is curious to investigate the way classical rules of chess are actually present in any move during the opening phase. Lastly, it is also designed for anyone who may want to explore a different way of studying opening theory, largely separated by the usual strategy of learning a certain

opening because "this is the main line"³

2 A brief overview of COT's design

In this section we will overview the central elements of COT's structure, as they have emerged during the project's planning phase. The steps, remarks and alterations will be presented in the chronological order they occurred during planning so as to let the path followed be revealed and allow for further remarks, corrections etc from third parties, so as to improve COT's design.

2.1 Learning strategy

The learning of COT will be completed in two layers in the way explained below, so as to achieve a better and more human-like performance.

Before we proceed, we mention that we will refer to a rule as *subjective* if it is dependent on the number of the move we are currently in either a non-probabilistic or a non-trivial⁴ way. On the contrary, we will consider a rule to be *objective* if its dependence on the number of the current move is either trivial or random. For instance, consider the following two rules:

R_1 : Do not move a piece twice!

R_2 : Do not make a move that leaves you with less pieces than the white!

The first rule, R_1 , is a subjective rule, since it is more probable that one will follow such a rule during the early phase of a game, when pieces should come out of their initial positions, rather than later on, when complex manoeuvres may be necessary, forcing a player to move one of their pieces for more than one or two consecutive moves. On the contrary, R_2 is an objective rule, in the sense that in almost all cases, regardless of the phase of the game, it is true that one should not sacrifice pieces for no obvious reason. Also, note that we decided to include randomness in objective and not subjective rules, interpreting a random behaviour of a rule's activation as a result of strong dependence on the chosen training examples and not on the strategical characteristics of chess opening.

2.1.1 The first layer — Learning subjective rules

In the first phase of our agent's learning procedure, we will focus on learning a predefined list of subjective rules from certain training examples chosen from past games played during the last 3 decades' world championships (see

³In a chess opening theory, the main line is the sequence of moves preferred against others during some specific opening because it statistically seems to offer more chances of win. For more, see [1].

⁴That is, it is not the constant function.

[2] and [3]). A sample list of these rules, but not an exhaustive one is the following:

- Move a different piece from the one moved previously!⁵
- Occupy more space (behind the pawns)!
- Control more space (using pieces' influence on each square)!
- Control the centre!
- Prepare/execute castling!
- Chekmate!
- ...

Our learning strategy will be as follows. Let $\mathcal{R} = \{R_1, \dots, R_n\}$ be a finite set of predefined rules, and let m be the number of training examples provided. We create an $n \times 10$ matrix⁶ $A = (a_{ij})$, which we update under the following protocol, for each training example, $e_1, \dots, e_m$:

We add 1 to a_{ij} iff rule R_i is being used during move j .

Then, we normalize our data in each column, dividing by the column sum, so as to get a probability distribution over the rules and over each column.

What the above procedure is actually doing, is letting our agent learn a linear function defined over:

$$\mathcal{R} \times \{1, 2, \dots, 10\},$$

which represents the probability that each rule is used in a specific move. Thus, we have a priority list of rules to use for each specific move.

2.1.2 The second layer — Learning objective rules

The main objective rule that will bother us is the following "equilibrium" rule:

Make a move so as to ensure $\geq$ piece value than the white!⁷

This is a general rule, however, a brief look at some of the most popular openings — e.g. the Queen's Gambit, see [1] — shows us that both the black and the white often sacrifice some piece (usually a pawn) to ensure some other strategical advantage — e.g. a quick attack, a faster development of the other pieces etc. So, under which criteria does a player decide to sacrifice a pawn so as to improve his position, in terms of strategical options? To answer

⁵We prefer to express all the rules in a positive way, avoiding "no" and "don't" expressions, so as to be easily comparable.

⁶We consider an opening to last, on average, 10 moves.

⁷We assume that we train our agent in playing with the black.

this question we shall first clarify the term of strategy in our framework. Strategy is, as we have mentioned, modelled by rules. However, not every rule may be considered to be a "strategic choice", since strategy depends on the current position. That is, the first category of subjective rules are the ones encoding strategy in terms of game-specific terms, while, the objective rule of keeping a close piece score with the opponent can be considered to be measure of "how far" we can go in the strategic side. That is, one may often decide to sacrifice a pawn to obtain some other benefits, however, a decision to sacrifice a second pawn is less likely to be made if the strategical merits of the new position are not that many and so on. Also, one may not sacrifice a pawn only to ensure that they control a4 square but they would sacrifice it so as to ensure a strong position at the center for their knight.

With the above in mind, while the piece-value equilibrium should be in general the goal, it is important for our agent to learn when to break this rule⁸ in favour of other, subjective, rules. To do so, we may note that in the above context, the equilibrium rule is a function of all the other rules and, more precisely, it is a function of the rules' state during a move. That is, one would not need to sacrifice a piece if the wanted strategic effect was present, but for its absence. So, it remains to re-study our training examples and, in the cases the equilibrium rule is broken, check which rules are *additionally* true in the case of the sacrifice, when compared to the case of no sacrifice. Our agent should learn both the rules themselves that occur, updating an $n \times 1$ array with the number of occurrences and the extend of the equilibrium break (qualitative data) as well as keep track of the number of rules that occur in each "equilibrium break", as a function of both the type of the rules and the already occurred ones, without the need of an equilibrium break (quantitative data). Thus, our agent is learning another function defined over $\mathcal{R}$ with $n + 1$ values⁹ (relative frequency of the rules plus the number of them).

2.1.3 Which move to chose, after all?

After having learned how to evaluate rules, it is high time we chose out move. We will use a strategy similar to restriction propagation to pick the "correct" move amongst the many available. To do so, we use the following algorithm:

⁸Either by sacrificing some piece or by proving willing to sacrifice one.

⁹One may raise a point of consistency with our definition of an objective rule versus a subjective rule here. More specifically, the equilibrium rule should be independent of the number of the current move, however, we express it as a function of the other, subjective, rules, which are directly dependent on the move count, resulting in equilibrium being indirectly dependent to it as well. Note, however, that what we express as a function of move count are the equilibrium rule *abandonments*, that is, we claim that while the equilibrium rule is a general fact for a chess game, whenever it may not be followed, it is due to certain strategic circumstances. Whether this consists a problem in our implementation or not, it remains to be verified.

For each $i=1,2,\dots,10$:

1. Prune the space of the available moves, choosing only these indicated by the rule R_j with the maximum probability of occurrence in column i .
2. Recursively repeat step 1 with the next rule, until the set of available moves is empty after the next repetition¹⁰.
3. Check for an equilibrium break. If the learnt values suggest that it should be made (e.g. the predicted rules are not activated without such a break), proceed with the break, else, pick a move randomly from the remaining moves without the equilibrium break¹¹.

Before we close this section, note how there is no direct rule to avoid checkmate. This is expected to be avoided due to the equilibrium rule, since in the usual piece rating system, the king is evaluated with ∞ , so it will be naturally avoided, if this rule is being followed correctly. Moreover, the cases in which a checkmate occurs in the first 10 moves (e.g. the Scholar's opening) are rare in the training examples' pool. However, should there emerge any need to handle such a situation, a checkmate analysis of 3 moves depth (e.g. using adversary search with an appropriate heuristic) may be used.

3 Further expansion of COT

Apart from the abovementioned main design of COR there are several extensions that could add to its learning abilities, some of them presented in the following paragraphs.

3.1 Search component

Even if the search option has been crossed out early on, since the intention of COT is to explore a rule-based approach of chess, there is a chance that a search component would boost COT's performance on certain aspects, such as checkmate check or a generalization of the COT's purposes, extending its range to the early middlegame, as well.

3.2 Introducing non-linear dependence between the rules

As we have seen, during the first layer, our agent learns a linear function over the rules and the move count, so as to decide which rule is more important at each stage of the opening. However, one could process training data in

¹⁰In this way, we have found the maximum number of rules that can be followed in this situation

¹¹Note again that such a break may be either a sacrifice or a sacrifice proposal — e.g. Queen's Gambit.

the same way, but for counting over *pairs* of rule or, in general, k -tuples of rules. A major drawback of such a strategy is its increased time complexity ($O(n^k)$ in terms of the number of the rules), however, it provides us with a bit of more insight, since the learnt function is non-linear, hence depicting more accurately the relation between the rules, the move count and the way they influence our general strategy.

However, this yields another problem. How would we propagate restrictions, when it comes to choose a move? One strategy would be, in the case of pairs, to pick the high frequency pair and use both rules in it, then take both rules and find another couple (not the same) that includes exactly one of them and has the highest frequency and continue in the same way, using the presented algorithm. The idea behind this way of rule selection is that, after having selected the highest rated pair for the current move, it is expected that a rule that is highly consistent with at least one of them, with at least one of them¹² is the most appropriate choice, providing a seemingly more coherent strategy.

Bibliography

- [1] <http://portablegamenotation.com/ECO.html>, retrieved on 29/03/2019.
- [2] Grivas, E. (2005), *Immortals: FIDE's World Championships, 1886-2005*, Kedros Publications.
- [3] <http://www.chessgames.com/index.html>, retrieved on 29/03/2019.

¹²We could, instead, choose the rule that is the most consistent with both rules, *on the average*.